Amirmohammad (Amir) Mohammadi

Computer Engineering PhD candidate | Texas A&M University

College Station, Texas | U.S. Permanent Resident

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Research interests: Machine learning, deep learning, computer vision, and AI for health; foundation models, multimodal learning, and physiological modeling.

Education

PhD in Computer Engineering · Texas A&M University

September 2022 – present · Expected to graduate in December 2026

PhD final examination passed in October 2026

Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing

MS in Electrical Engineering · Sharif University of Technology

September 2018 – February 2021

BS in Electrical Engineering · University of Tabriz

September 2014 – September 2018

Research experience

Graduate Research Assistant · Texas A&M University

September 2022 – present

- Developed parameter-efficient fine-tuning (PEFT) methods for Transformer foundation models, adapting pretrained representations to new classification tasks through distribution-aware modules.
- Built a few-shot histopathology classification pipeline combining vision-language models, attention-based multiple instance learning, and visual and textual adaptation.
- Improved mean balanced accuracy from 43.5% to 78.1% in a four-shot whole-slide classification study using subject-disjoint splits and repeated evaluation.
- Contributed to a modular evaluation pipeline for image classification, semantic segmentation, and object detection, comparing pretrained computer vision models and adaptation strategies.
- Developed physics-informed neural networks for physiological time-series prediction; the collaborative blood-pressure study reduced ground-truth training requirements by an average factor of 15.
- Benchmarked cross-domain transfer learning, finding a 3-percentage-point classification accuracy improvement from image-pretrained models over audio-pretrained models.

Graduate Student Researcher · Sharif University of Technology

July 2019 – February 2021

- Designed an integrated ECG/EDA wearable sensor, from circuitry and Bluetooth Low Energy electronics to machine-learning algorithms.
- Achieved 94% stress-detection accuracy in an 18-participant study and threefold battery life relative to the evaluated alternatives.

Technical skills

Programming & scientific computing: Python, C, MATLAB, NumPy, pandas, Matplotlib

Machine learning frameworks: PyTorch, Hugging Face Transformers, scikit-learn

Development tools: Git, Linux, Jupyter Notebook, high-performance computing

Publications

Peer-reviewed & accepted publications

J. Ritu, **A. Mohammadi**, D. Carreiro, A. Van Dine, J. Peeples. **Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification**. *IEEE Access (2026)*. Preprint · PDF

A. Mohammadi, D. Carreiro, A. Van Dine, J. Peeples. **Histogram-based Parameter-efficient Tuning for Passive and Active Sonar Classification**. *IEEE International Geoscience and Remote Sensing Symposium (IGARSS) (2026) · Accepted*. Preprint · PDF · Code

F. Orvati Nia, **A. Mohammadi**, S. Al Kharsa, P. Naikare, Z. Hampel-Arias, J. Peeples. **Neighborhood Feature Pooling for Remote Sensing Image Classification**. *IEEE/CVF WACV Workshops (2026), pp. 1393–1402*. Paper

A. Mohammadi, I. Masabarakiza, E. Barnes, D. Carreiro, A. Van Dine, J. Peeples. **Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks**. *OCEANS 2025 Brest (2025), pp. 1–6*. Publisher

A. Mohammadi, T. Kelhe, D. Carreiro, A. Van Dine, J. Peeples. **Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models**. *OCEANS 2025 Brest (2025), pp. 1–6*. Publisher

K. Sel, **A. Mohammadi**, R. I. Pettigrew, R. Jafari. **Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation**. *npj Digital Medicine (2023), vol. 6, article 110*. Paper · PDF

A. Mohammadi, M. Fakharzadeh, B. Baraeinejad. **An Integrated Human Stress Detection Sensor Using Supervised Algorithms**. *IEEE Sensors Journal (2022), vol. 22, no. 8, pp. 8216–8223*. Publisher

Manuscripts under review

A. Mohammadi, J. Peeples, A. Van Dine. **Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification**. *arXiv:2606.02341 (2026) · Under review*. Preprint · PDF

A. Mohammadi, R. Khadka, K. A. Grant, M. L. Benton, J. Peeples. **Predicting Ethanol Drinking Status from Nonhuman Primate Liver Histology: A Proof of Concept for Parameter-efficient Fine-tuning in Few-Shot Vision-Language Modeling**. *Manuscript (2026) · Under review*.

Academic service, teaching & mentoring

Reviewer, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). Reviewer, Expert Systems with Applications. Session chair, IEEE International Geoscience and Remote Sensing Symposium (IGARSS).

Mentored undergraduate and graduate students on end-to-end AI projects. Delivered guest lectures in Machine Learning for Electrical Engineering at Texas A&M University. Showcased AI projects at Night at the Zach during the Texas Science & Engineering Fair.

Awards & recognition

IEEE Geoscience and Remote Sensing Society travel grant · May 2026

Texas A&M Electrical & Computer Engineering conference travel grant · May 2026

Fully funded doctoral research assistantship support from MIT Lincoln Laboratory, the National Institutes of Health, and the Texas A&M Department of Electrical & Computer Engineering.